

LINEAR SEARCH:

Linear Search is the simplest searching algorithm. It searches for an element by checking each element of the array one by one from the beginning until the desired element is found or the end of the array is reached.

Algorithm:

1. Read the number of elements.
2. Enter the array elements.
3. Read the element to be searched.
4. Start from the first element of the array.
5. Compare each element with the target element.
6. If a match is found, display its index or position.
7. If the end of the array is reached without finding the element, display "Element not found."

Output:

```
Enter the number of elements:
5
Enter the elements:
78
65
98
23
4
enter the element to search:
4
Element found at index :4
Process finished with exit code 0
```

Time Complexity:

Case	Time Complexity
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Best Case	$O(1)$
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Average Case	$O(n)$
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Worst Case	$O(n)$
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Space Complexity:

Type	Space Complexity
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Linear Search	$O(1)$
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